

GAUTAM BUDDHA UNIVERSITY



SCHOOL OF ICT

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DETAILED SYLLABI (SEMESTER III-IV)

M.TECH. (COMPUTER SCIENCE AND ENGINEERING)

SPECIALIZATION IN

ARTIFICIAL INTELLIGENCE

ACADEMIC BATCH: 2025-2027

SEMESTER III



Cloud and Edge Computing			
Course Code:	CS621	Course Credits:	3
Course Category:	PCC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Explain the principles and terminology of the foundations of Cloud and Edge Computing.			
2. Analyse problems and trade-offs concerning cloud architecture and virtualization.			
3. Apply suitable methods drawn from distributed data and compute.			
4. Design solutions that integrate edge-cloud continuum.			
5. Evaluate approaches and responsible practice relating to security, reliability and operations.			
COURSE OUTCOMES			
At the end of the course, the students should be able to:			
1. Explain the concepts and terminology of the foundations of Cloud and Edge Computing			
2. Analyse problems using the methods covered under cloud architecture and virtualization.			
3. Apply appropriate techniques from distributed data and compute to realistic datasets or systems.			
4. Design a solution that integrates concepts from the edge-cloud continuum.			
5. Evaluate alternative approaches using technical, performance and reliability criteria.			
6. Develop and communicate a responsible, reproducible solution drawing on security, reliability and operations.			

UNIT I: FOUNDATIONS OF DISTRIBUTED COMPUTING

Evolution from clusters and grids to cloud and edge computing; Definition and essential characteristics of cloud computing — on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service; Overview of service models (IaaS, PaaS, SaaS) and deployment models (public, private, hybrid, multi-cloud); Drivers for edge computing — latency, bandwidth, privacy and data locality; The cloud-to-edge continuum at a glance; Representative cloud and edge use cases; Benefits, limitations and cost/performance trade-offs.

UNIT II: CLOUD ARCHITECTURE AND VIRTUALIZATION

IaaS, PaaS and SaaS; Virtual machines and containers; Orchestration; Service discovery; Autoscaling; Multi-tenancy; Cloud storage; Hypervisor types; Full and para-virtualization; Hardware support; Namespaces and control groups; Container images and registries; VM-container comparison; Orchestration control loops; Resource isolation and scheduling; Public, private, hybrid and multi-cloud models; Regions and availability zones; Reference architectures; Elasticity policies; Resource provisioning; Load balancing; Cloud service selection; Performance, availability and cost comparison; Worked examples and case analysis.

UNIT III: DISTRIBUTED DATA AND COMPUTE

Distributed file systems; Key-value and NoSQL stores; Partitioning and replication; MapReduce, stream processing and serverless workflows; Data locality; Shuffling and partition strategies; Distributed joins; Lineage and recomputation; Batch-stream unification; Data skew; Checkpointing; Back-pressure; Execution-plan analysis; Scalability experiments; Single-writer and multi-writer replication; Leader-follower and leaderless designs; Quorum reads and writes; Conflict-free replicated data types; Consistency anomalies; Isolation levels; Compensating actions; Failure recovery; Worked examples and case analysis; Comparative evaluation of techniques.

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UNIT IV: EDGE-CLOUD CONTINUUM

Fog and edge architectures; Placement and offloading; Latency, bandwidth and energy trade-offs; IoT integration; Federated edge learning; Public, private, hybrid and multi-cloud models; Regions and availability zones; Reference architectures; Elasticity policies; Resource provisioning; Load balancing; Cloud service selection; Performance, availability and cost comparison; Edge node taxonomy; Cloudlets and gateways; Task-placement formulations; Computation offloading; Mobility support; Intermittent connectivity; Edge caching; Privacy-preserving analytics; Latency-energy modelling; Smart-city and industrial-IoT examples; Worked examples and case analysis.

UNIT V: SECURITY, RELIABILITY AND OPERATIONS

Zero-trust basics; Identity and access; Observability; SRE indicators; Resilience patterns; Cost governance; Sustainable computing; Case studies; Availability and reliability measures; Redundancy; Health checks; Graceful degradation; Bulkheads; Recovery-point and recovery-time objectives; Backup and disaster recovery; Fault injection; Chaos experiments; Reliability-cost trade-offs; Logs, metrics and traces; Service-level indicators, objectives and error budgets; Capacity planning; Deployment strategies; Runbooks; Incident response; Root-cause analysis; Post-incident reviews; Operational dashboards; Worked examples and case analysis.

Text Books/Reference Books:

1. M. van Steen and A. S. Tanenbaum, Distributed Systems, 4th ed., distributed-systems.net, 2023.
2. Erl, R. Puttini and Z. Mahmood, Cloud Computing: Concepts, Technology & Architecture, Pearson.
3. T. Hwang, G. Fox and J. Dongarra, Distributed and Cloud Computing, Morgan Kaufmann.
4. R. Buyya and A. V. Dastjerdi (eds.), Internet of Things: Principles and Paradigms, Morgan Kaufmann. C. Pahl et al., Cloud Container Technologies: a State-of-the-Art Review, IEEE Transactions on Cloud Computing.

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Soft Computing Techniques			
Course Code:	CA623	Course Credits:	3
Course Category:	PCC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. To expose the concepts of feedforward neural networks.			
2. To provide adequate knowledge about feedback neural networks.			
3. To teach about the concept of fuzziness involved in various systems.			
4. To expose the ideas about genetic algorithms.			
5. To provide adequate knowledge about FLC and NN toolbox.			
COURSE OUTCOMES			
At the end of the course, the students should be able to:			
1. Learn and understand Fuzzy logic and its applications.			
2. Comprehend Artificial Neural Networks and their applications.			
3. Solve single-objective optimization problems using GAs.			
4. Solve multi-objective optimization problems using Evolutionary algorithms.			
5. Applications of Soft Computing to solve problems in a variety of application domains.			
6. Assess Fuzzy logic and its components effectively.			

UNIT I: INTRODUCTION

Introduction to Soft Computing; Definition; Requirement; Necessity; Adequacy; Various dialects of Soft Computing; Evolutionary Algorithms; Fuzzy Sets and Fuzzy Logic; Artificial Neural Networks – Their suitability in searching, optimization, decision making and pattern related problems; Potential areas of applications.

UNIT II: FUZZY SETS AND FUZZY LOGIC

Introduction to Fuzzy Sets and Fuzzy Logic; Difference between classical and fuzzy sets; Chance versus fuzziness; Limitations of fuzzy systems; Typical shapes of membership functions and their usage; Operations on fuzzy sets: Complement, Intersection, Union; Combinations on operations; Aggregation operation.

UNIT III: FUZZY RELATIONS AND FUZZY SYSTEMS

Cartesian Product; Classical Relations and Fuzzy Relations; Cardinality; Operations and properties of crisp and fuzzy relations; Composition of operations; Fuzzy Cartesian Product; The linguistic variables; Reasoning in fuzzy logic; Fuzzification and Defuzzification; Mamdani and Sugeno Fuzzy Inference Systems.

UNIT IV: NEURAL NETWORK

Overview of biological neurons; McCulloch–Pitts model; Rosenblatt's Perceptron model; Difference, capabilities and limitations; Model of generic computational neuron; Basic activation functions; Basic learning laws of neurons; Single-layer and multilayer architectures; Feedforward and feedback networks.

UNIT V: LEARNING FUNDAMENTALS

Problems suitable and not suitable for applying evolutionary algorithms; Various dialects of evolutionary algorithms; Terminology of Genetic Algorithms; Canonical Genetic Algorithm; Common

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representations and related reproduction operators; Premature convergence; Schema theorem; Minimal deceptive problem and Royal Road function; Fitness function; Roulette wheel selection; Rank selection; Tournament Selection; Termination criteria; Survivor selection; Population models; Parallel implementation.

Text Books / Reference books:

1. Artificial Neural Networks: An Introduction to ANN Theory and Practice, Peteus J. Braspenning, PHI Publication, 2005.
2. Fuzzy Logic: A Spectrum of Theoretical and Practical Issues, Paul P. Wang, Pearson Publication, 2004.
3. An Introduction to Genetic Algorithms, Melanie Mitchell, MIT Press, 1998.
4. A Genetic Algorithm Tutorial, Darrell Whitley.
5. Fuzzy Sets, Fuzzy Logic, and Fuzzy Systems: Selected Papers, Lotfi Asker Zadeh, George J. Klir, Bo Yuan, 2005.
6. Foundations of Fuzzy Logic and Soft Computing, 12th International Fuzzy Conference Proceedings, 2005.
7. Neural Networks: Theory, Practice and Applications, Martin Anthony, Oxford University Press, 2003.
8. Neural Networks Theory and Application, Oscar Castillo, Wiley Eastern Publication.
9. Genetic Algorithms in Search, Optimization and Machine Learning, David E. Goldberg, Addison-Wesley, 1988.

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Soft Computing Techniques Lab			
Course Code:	CA681	Course Credits:	2
Course Category:	PCC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Labs (Hrs/Week):	1 (4 HRS)	Mid Sem. Exam Hours:	-
Total No. of Labs:	10	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Provide an understanding of the basic mathematical elements of fuzzy sets.			
2. To use fuzzy set operations to implement current computing techniques used in fuzzy computing			
3. To implement applications using fuzzy set operations.			
4. Cover fuzzy logic inference with emphasis on their use in the design of intelligent or humanistic systems.			
5. To become familiar with neural network learning algorithms from available examples.			
COURSE OUTCOMES			
At the end of the course, the students should be able to:			
1. Analyse applications using fuzzy set which uses current techniques, skills, and tools necessary for computing.			
2. To identify and solve engineering problems using the fuzzy set theory and identify the differences and similarities between fuzzy sets and classical set theories			
3. Learn the mathematical basis as well as the general principles of various soft computing techniques			
4. Become familiar with fuzzy relations and the properties of these relations			
5. Identify, analyse, design, and solve the problem, implement, and validate the solution including both hardware and software.			

PRACTICALS

Note: Each practical shall include an objective, design/algorithm, implementation, test data, observations, results and a brief critical analysis. Faculty may adapt platforms while retaining the stated learning outcomes.

1. Implementation of operations and properties of classical sets.
2. Implementation of operations and properties of Fuzzy Sets.
3. Implementation of a Cartesian Product and Fuzzy Cartesian Product.
4. Implementation of composition of classical and fuzzy relations.
5. Implementing a simple perceptron model to show the effect of inputs, weights, and bias on the decision boundary.
6. Train the simple perceptron to show its limitation to solve only linearly separable boundary problems.
7. To design, implement and train a feedforward ANN to solve a pattern recognition problem.
8. Implementation of an adaptive filter using ANN.
9. Implementing SGA to solve a function optimization problem in a given range.
10. Implementation of genetic operators for Permutation Representation.
11. Implementation of Roulette Wheel Selection.
12. Implementation of Stochastic Universal Sampling.
13. Implementation of Rank Selection.
14. Implementation of Tournament Selection.
15. Illustrate the effect of Sigma Scaling.

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PROGRAM ELECTIVES

Prof. Dr. P. S. S. S.

AI for Drone Systems			
Course Code:	CA631	Course Credits:	3
Course Category:	PEC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Explain the fundamentals of Artificial Intelligence in autonomous drone systems.			
2. Analyse AI techniques for drone perception, navigation, and decision-making.			
3. Apply computer vision, machine learning, and deep learning for intelligent aerial applications.			
4. Design autonomous drone solutions integrating perception, planning, and edge AI.			
5. Evaluate AI-enabled drone systems considering performance, safety, reliability, and regulations.			
COURSE OUTCOMES			
At the end of the course, the students should be able to:			
1. Explain AI concepts and architectures used in autonomous drone systems.			
2. Analyze perception, localization, and navigation techniques for drones.			
3. Apply machine learning and deep learning algorithms to drone applications.			
4. Design intelligent drone solutions for real-world autonomous missions.			
5. Evaluate AI-based drone systems using technical, operational, and safety metrics.			
6. Develop AI-enabled drone applications considering ethics, privacy, security, and regulatory requirements.			

UNIT I: INTRODUCTION TO AI-ENABLED DRONE SYSTEMS

Drone platforms and architectures; UAV classifications; Flight dynamics fundamentals; Sensors and actuators; GPS, IMU and LiDAR; Drone communication systems; Embedded AI; AI-enabled autonomy; Mission planning; Drone operating modes; System architecture; Drone applications; Drone software stack; AI frameworks for UAVs; Performance measures; Safety considerations; Contemporary drone platforms; Industry case studies.

UNIT II: COMPUTER VISION FOR DRONES

Image acquisition; Camera calibration; Image preprocessing; Feature extraction; Object detection; Image segmentation; Object tracking; Optical flow; Stereo vision; Depth estimation; Pose estimation; Visual odometry; Visual SLAM; Semantic segmentation; Scene understanding; Obstacle detection; Landing zone detection; Aerial image analysis; Remote sensing; Deep learning for aerial vision; YOLO; Vision Transformers; Comparative evaluation of techniques; Performance measures.

UNIT III: AI FOR AUTONOMOUS NAVIGATION

Localization; Mapping; Path planning; Motion planning; Trajectory optimization; Obstacle avoidance; Simultaneous Localization and Mapping (SLAM); Reinforcement Learning; Decision making; Autonomous waypoint navigation; Mission optimization; Multi-objective optimization; Adaptive navigation; Sensor fusion; Kalman Filter; Particle Filter; Deep Reinforcement Learning; Dynamic path replanning; Swarm intelligence; Multi-drone coordination; Comparative evaluation of techniques; Industry applications.

UNIT IV: EDGE AI AND INTELLIGENT DRONE APPLICATIONS

Edge AI; TinyML; Onboard inference; Model optimization; Model compression; Quantization; TensorRT; NVIDIA Jetson; Qualcomm RB5; Embedded GPU computing; Real-time analytics; Aerial surveillance; Precision agriculture; Infrastructure inspection; Disaster management; Environmental

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monitoring; Smart city applications; Healthcare delivery; Logistics and delivery drones; Energy-efficient AI; Comparative evaluation of techniques; Performance optimization; Contemporary tools and practices.

UNIT V: SAFETY, SECURITY AND FUTURE DIRECTIONS

AI safety; Fault detection; Predictive maintenance; Explainable AI; Responsible AI; Privacy-preserving AI; Secure drone communication; Cybersecurity threats; Adversarial attacks; Federated learning; Digital twins; Cloud robotics; Regulatory frameworks; DGCA regulations; Airspace management; UTM systems; Ethical AI; Sustainable drone systems; AI governance; Emerging research trends; Industry case studies; Scalability considerations.

Textbooks / Reference books:

1. Autonomous Flying Robots – Kenzo Nonami, Springer.
2. Programming Autonomous Drones with ROS 2 – Packt Publishing.
3. Artificial Intelligence for Robotics – Robin Murphy, MIT Press.
4. Computer Vision: Algorithms and Applications – Richard Szeliski.
5. Deep Learning – Ian Goodfellow, Yoshua Bengio, Aaron Courville.
6. Small Unmanned Aircraft: Theory and Practice – Randy Beard and Timothy McLain.

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Autonomous Mobile Robots			
Course Code:	CA633	Course Credits:	3
Course Category:	PEC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Explain the principles and terminology of mobile robot models.			
2. Analyse problems and trade-offs concerning localization.			
3. Apply suitable methods drawn from mapping and SLAM.			
4. Design solutions that integrate planning and navigation.			
5. Evaluate approaches and responsible practice relating to system integration.			
COURSE OUTCOMES			
At the end of the course, the students should be able to:			
1. Explain the fundamental concepts and terminology of mobile robot models.			
2. Analyse problems using the methods covered under localization.			
3. Apply appropriate techniques from mapping and SLAM to realistic datasets or systems.			
4. Design a solution that integrates concepts from planning and navigation.			
5. Evaluate alternative approaches using technical, performance and reliability criteria.			
6. Develop and communicate a responsible, reproducible solution drawing on system integration.			

UNIT I: MOBILE ROBOT MODELS

Locomotion; Wheeled configurations; Kinematic and dynamic models; Motion constraints; Differential-drive, tricycle, car-like and omnidirectional robots; Wheel constraints; Instantaneous centre of rotation; Forward and inverse kinematics; Non-holonomic constraints; Odometry; Dynamic effects; Controllability; Calibration; Trajectory feasibility; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations; Simulation-based exercises; Hardware deployment issues.

UNIT II: LOCALIZATION

Odometry; Probabilistic motion and sensor models; Kalman, histogram and particle-filter localization; Motion and observation models; Dead reckoning; Landmark association; Covariance propagation; EKF, UKF and particle filtering; Global localization; Kidnapped-robot problem; Accuracy and consistency measures; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations; Simulation-based exercises; Hardware deployment issues.

UNIT III: MAPPING AND SLAM

Occupancy grids; Feature maps; Graph-based SLAM; Loop closure; Map quality; Inverse sensor models; Scan matching; ICP; Visual and LiDAR odometry; EKF-SLAM; Particle-filter SLAM; Pose graphs; Loop-closure detection; Graph optimization; Map consistency and quality evaluation; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations; Simulation-based exercises; Hardware deployment issues.

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UNIT IV: PLANNING AND NAVIGATION

Global and local planning; Costmaps; Reactive methods; Dynamic obstacles; Exploration; Problem representation; Search trees and graphs; Dijkstra and A*; Admissible and consistent heuristics; Potential fields; Dynamic window; RRT and RRT*; Trajectory smoothing; Replanning; Completeness and optimality; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations; Simulation-based exercises.

UNIT V: SYSTEM INTEGRATION

Navigation stacks; Multi-robot systems; Field robotics; Benchmarking; Safety and deployment case studies; Sensor, compute and actuator interfaces; Middleware timing; State-estimation pipeline; Navigation-stack configuration; Diagnostics; Logging and replay; Simulation-to-hardware transfer; Safety interlocks; Acceptance testing; Field benchmarking; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations; Simulation-based exercises.

Textbooks / Reference books:

1. R. Siegwart, I. Nourbakhsh and D. Scaramuzza, Introduction to Autonomous Mobile Robots, 2nd ed., MIT Press.
2. S. Thrun, W. Burgard and D. Fox, Probabilistic Robotics, MIT Press.
3. S. M. LaValle, Planning Algorithms, Cambridge.
4. P. Corke, Robotics, Vision and Control, 3rd ed., Springer.
5. ROS 2 and Nav2 official documentation.

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Agentic AI			
Course Code:	CA635	Course Credits:	3
Course Category:	PEC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3

COURSE OBJECTIVES

1. Explain the principles and terminology of agentic artificial intelligence and autonomous intelligent agents.
2. Analyse problems and trade-offs concerning planning, reasoning and decision-making in agentic systems.
3. Apply suitable methods drawn from large language models, tool integration and multi-agent collaboration.
4. Design agentic AI systems for solving real-world autonomous tasks.
5. Evaluate approaches and responsible practice relating to autonomous AI agents.

COURSE OUTCOMES

At the end of the course, the students should be able to:

1. Explain the fundamental concepts and terminology of agentic AI and intelligent agents.
2. Analyse problems using the methods covered under planning, reasoning and autonomous decision-making.
3. Apply appropriate techniques for tool usage, workflow automation and multi-agent collaboration.
4. Design an agentic AI solution for a specified real-world application.
5. Evaluate alternative agentic AI approaches using technical, performance and reliability criteria.
6. Develop and communicate a responsible, reproducible agentic AI solution.

UNIT I: FOUNDATIONS OF AGENTIC AI

Intelligent agents; Agentic AI fundamentals; Characteristics of autonomous agents; Agent architectures; Perception, reasoning and action; Reactive and deliberative agents; Goal-oriented behaviour; Planning concepts; Memory and context management; Agent life cycle; Applications; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations.

UNIT II: AGENT REASONING AND TOOL INTEGRATION

Reasoning strategies; Task decomposition; Planning and execution; Tool calling; API integration; External knowledge retrieval; Workflow automation; Memory management; Agent orchestration; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations; Simulation-based exercises.

UNIT III: MULTI-AGENT SYSTEMS

Multi-agent systems; Collaborative agents; Communication protocols; Coordination strategies; Task allocation; Negotiation; Conflict resolution; Distributed problem solving; Human-agent collaboration; Agent evaluation; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations; Reproducibility practices.

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UNIT IV: AGENTIC AI FRAMEWORKS AND APPLICATIONS

Agent development frameworks; Workflow design; Conversational agents; Coding assistants; Autonomous research assistants; Enterprise automation; Robotics applications; Software engineering applications; Decision support systems; Healthcare and finance use cases; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations.

UNIT V: DEPLOYMENT, EVALUATION AND RESPONSIBLE AGENTIC AI

Agent evaluation; Benchmarking; Reliability; Scalability; Monitoring; Explainability; Human oversight; Fairness; Bias; Privacy; Security; Ethical and legal considerations; Governance; Responsible AI; Emerging trends in Agentic AI; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance measures; Common limitations and failure scenarios; Future research directions.

Textbooks / Reference books:

1. Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, 4th Edition, Pearson.
2. Wooldridge, M., An Introduction to MultiAgent Systems, 2nd Edition, Wiley.
3. Bratman, M., Intentions, Plans, and Practical Reason, Harvard University Press.
4. Foster, D., Generative Deep Learning, 2nd Edition, O'Reilly Media.
5. Attention mechanisms. Luger, G. F., Artificial Intelligence: Structures and Strategies for Complex Problem Solving, 8th Edition,



Intelligent Information Retrieval			
Course Code:	CA637	Course Credits:	3
Course Category:	PEC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Explain the principles and architecture of intelligent information retrieval systems.			
2. Analyse indexing, ranking, and retrieval techniques for structured and unstructured data.			
3. Apply machine learning and deep learning techniques for intelligent search.			
4. Design retrieval systems using semantic search, knowledge graphs, and Retrieval-Augmented Generation (RAG).			
5. Evaluate intelligent retrieval systems using effectiveness, efficiency, and user-centric metrics.			
COURSE OUTCOMES			
At the end of the course, the students should be able to:			
1. Explain the concepts and architecture of intelligent information retrieval systems.			
2. Analyse indexing, ranking, and query processing techniques for information retrieval.			
3. Apply machine learning and deep learning methods for intelligent search applications.			
4. Design semantic retrieval systems integrating embeddings, knowledge graphs, and RAG.			
5. Evaluate retrieval models using effectiveness, efficiency, scalability, and relevance metrics.			
6. Develop intelligent retrieval solutions using contemporary AI frameworks and responsible AI practices.			

UNIT I: FOUNDATIONS OF INFORMATION RETRIEVAL

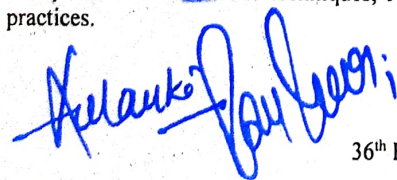
Information retrieval fundamentals; Information retrieval models; Structured, semi-structured and unstructured data; Document representation; Tokenization; Text normalization; Stop-word removal; Stemming and lemmatization; Inverted index; Positional index; Boolean retrieval; Vector Space Model; TF-IDF; BM25 ranking; Query processing; Query expansion; Relevance feedback; Retrieval architecture; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures.

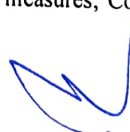
UNIT II: MACHINE LEARNING FOR INFORMATION RETRIEVAL

Learning-to-rank; Feature engineering; Ranking functions; Query classification; Document classification; Text clustering; Recommendation techniques; Personalised search; Collaborative filtering; Content-based filtering; Search log analysis; Click-through prediction; User behaviour modelling; Search analytics; Learning-to-rank algorithms; Supervised and unsupervised retrieval; Comparative evaluation of techniques; Performance measures; Contemporary tools and practices; Scalability considerations; Industry case studies.

UNIT III: DEEP LEARNING AND SEMANTIC SEARCH

Word embeddings; FastText; Word2Vec; GloVe; Contextual embeddings; BERT; Sentence Transformers; Dense retrieval; Dual encoders; Cross encoders; Semantic search; Neural ranking; Vector databases; Approximate Nearest Neighbor (ANN) search; FAISS; Pinecone; ChromaDB; Hybrid retrieval; Cross-lingual retrieval; Multimodal retrieval; Worked examples and case analysis; Comparative evaluation of techniques; Performance and quality measures; Contemporary tools and practices.




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UNIT IV: KNOWLEDGE-AWARE RETRIEVAL AND GENERATIVE AI

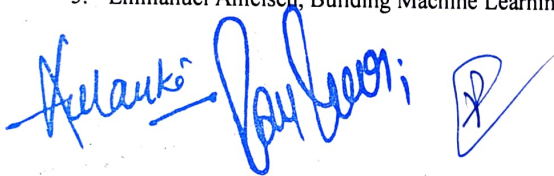
Knowledge graphs; Entity linking; Entity retrieval; Ontologies; SPARQL; Graph; Graph neural networks for retrieval; Retrieval-Augmented Generation (RAG); Vector search; Hybrid search; agentic AI for retrieval; Query rewriting; Document chunking; Embedding generation; Context retrieval; Hallucination mitigation; Citation generation; Enterprise search; AI-powered question answering; Comparative evaluation of techniques; Contemporary AI tools and practices; Responsible AI considerations.

UNIT V: EVALUATION, DEPLOYMENT AND APPLICATIONS

Evaluation metrics; Precision; Recall; MAP; MRR; NDCG; User satisfaction; Search latency; Scalability; Distributed indexing; Cloud-based retrieval; Edge search; Privacy-preserving retrieval; Explainable retrieval; Bias and fairness in search; Secure retrieval; Retrieval monitoring; Model deployment; Search APIs; Enterprise search systems; Web search; Academic search; Healthcare retrieval; Legal information retrieval; Recommendation systems; Worked examples and case analysis; Contemporary tools and practices; Industry case studies.

Textbooks / Reference books:

1. Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze, Introduction to Information Retrieval, Cambridge University Press.
2. Ricardo Baeza-Yates and Berthier Ribeiro-Neto, Modern Information Retrieval, Addison-Wesley.
3. Stefan Büttcher, Charles L. A. Clarke, and Gordon V. Cormack, Information Retrieval: Implementing and Evaluating Search Engines, MIT Press.
4. Tommaso Teofili, AI-Powered Search, Manning Publications.
5. Emmanuel Ameisen, Building Machine Learning-Powered Applications, O'Reilly Media.



Introduction to Large Language Models			
Course Code:	CA639	Course Credits:	3
Course Category:	PEC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3

COURSE OBJECTIVES

1. Explain the principles and terminology of large language models and transformer-based architectures.
2. Analyse problems and trade-offs concerning language modelling, prompting and model adaptation.
3. Apply suitable methods drawn from prompt engineering, retrieval augmentation and fine-tuning.
4. Design LLM-based solutions for real-world natural language processing applications.
5. Evaluate approaches and responsible practice relating to large language models.

COURSE OUTCOMES

At the end of the course, the students should be able to:

1. Explain the fundamental concepts and terminology of large language models and transformer architectures.
2. Analyse problems using the methods covered under language modelling and prompt engineering.
3. Apply appropriate techniques for prompting, retrieval augmentation and model adaptation to realistic applications.
4. Design an LLM-based solution for a specified natural language processing task.
5. Evaluate alternative LLM approaches using technical, performance and reliability criteria.
6. Develop and communicate a responsible, reproducible LLM-based solution.

UNIT I: FOUNDATIONS OF LARGE LANGUAGE MODELS

Natural language processing fundamentals; Evolution of language models; Statistical language models; Neural language models; Transformer architecture; Tokenization; Attention mechanism; Positional encoding; Pre-training concepts; Foundation models; Applications of LLMs; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations.

UNIT II: LLM ADAPTATION AND CONTEXT MANAGEMENT

Prompting fundamentals for LLM interaction; Zero-shot, one-shot, and few-shot learning; Basic instruction and structured prompting; Context-window management; Retrieval-augmented generation fundamentals; Fine-tuning concepts; Parameter-efficient adaptation; Model alignment and instruction tuning; Model limitations and adaptation trade-offs; Evaluation of adapted LLMs.

UNIT III: LLM APPLICATIONS

Text generation; Summarization; Question answering; Conversational AI; Information extraction; Text classification; Code generation; Translation; Document analysis; Enterprise applications; Domain-specific LLMs; Multimodal LLM overview; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations; Reproducibility practices.

UNIT IV: LLM DEPLOYMENT AND INTEGRATION

LLM inference; APIs and cloud-based LLM services; Open-source LLMs; Model serving; Workflow automation; Tool calling; Function calling; AI assistants; Integration with external knowledge sources; Performance optimization; Monitoring and evaluation; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations.

UNIT V: RESPONSIBLE LLMs AND EMERGING TRENDS

Hallucinations; Explainability; Bias and fairness; Privacy; Security; Copyright and intellectual property; Ethical considerations; Responsible AI; LLM evaluation benchmarks; Governance; Model alignment; Emerging trends in LLMs; Agentic AI overview; Future directions; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance measures; Common limitations and failure scenarios; Contemporary tools and practices; Ethical and societal implications.

Textbooks / Reference books:

1. Lewis Tunstall, Leandro von Werra and Thomas Wolf, Natural Language Processing with Transformers, O'Reilly Media.
2. Denis Rothman, Transformers for Natural Language Processing, 3rd Edition, Packt Publishing.
3. Jacob Eisenstein, Natural Language Processing, MIT Press.
4. Yoav Goldberg, Neural Network Methods for Natural Language Processing, Morgan & Claypool Publishers.
5. Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, 4th Edition, Pearson.



AI Planning and Decision Making			
Course Code:	CA641	Course Credits:	3
Course Category:	PEC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Explain the principles and terminology of state-space planning.			
2. Analyse problems and trade-offs concerning classical planning.			
3. Apply suitable methods drawn from planning under uncertainty.			
4. Design solutions that integrate temporal and multi-agent planning.			
5. Evaluate approaches and responsible practice relating to integrated decision systems.			
COURSE OUTCOMES			
At the end of the course, the students should be able to:			
1. Explain the fundamental concepts and terminology of state-space planning.			
2. Analyse problems using the methods covered under classical planning.			
3. Apply appropriate techniques from planning under uncertainty to realistic datasets or systems.			
4. Design a solution that integrates concepts from temporal and multi-agent planning.			
5. Evaluate alternative approaches using technical, performance and reliability criteria.			
6. Develop and communicate a responsible, reproducible solution drawing on integrated decision systems.			

UNIT I: STATE-SPACE PLANNING

Problem representation; Uninformed and heuristic search; Admissibility; Planning graphs; Problem states, actions, goals and costs; Uninformed search; Uniform-cost search; Greedy best-first and A-star; Admissible and consistent heuristics; Graph-search implementation; Duplicate detection; Memory-bounded search; Complexity and optimality; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations; Simulation-based exercises.

UNIT II: CLASSICAL PLANNING

STRIPS/PDDL; Forward and backward planning; Partial-order planning; Heuristics; STRIPS and PDDL representation; Preconditions and effects; Planning graphs; Delete relaxation; Heuristic construction; Landmark methods; Plan validation; Benchmark domains; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations; Simulation-based exercises; Hardware deployment issues; Reproducibility practices; Ethical and societal implications.

UNIT III: PLANNING UNDER UNCERTAINTY

Bayesian decision theory; MDPs; Value and policy iteration; POMDPs; Belief states; Markov decision processes; Bellman equations; Partially observable MDPs; Belief update; Approximate planning; Risk and uncertainty measures; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations; Simulation-based exercises; Hardware deployment issues; Reproducibility practices; Ethical and societal implications.

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UNIT IV: TEMPORAL AND MULTI-AGENT PLANNING

Time and resources; Scheduling; Decentralized planning; Game-theoretic reasoning; Durative actions; Temporal constraints; Resources; Planning with deadlines; Coordination protocols; Negotiation; Conflict resolution; Multi-agent plan evaluation; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations; Simulation-based exercises; Hardware deployment issues; Reproducibility practices; Ethical and societal implications; Recent research directions.

UNIT V: INTEGRATED DECISION SYSTEMS

Task and motion planning; Online/receding-horizon planning; Learning-guided planning; Safety and explainability; Task-and-motion planning; Receding-horizon and model-predictive decisions; Planning-learning interfaces; Execution monitoring; Contingency handling; Recovery; Resource constraints; Explainable decisions; Safety constraints; Scenario-based evaluation; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations; Simulation-based exercises; Hardware deployment issues.

Textbooks / Reference books:

1. S. Russell and P. Norvig, Artificial Intelligence: A Modern Approach, 4th ed., Pearson.
2. M. Ghallab, D. Nau and P. Traverso, Automated Planning and Acting, Cambridge.
3. S. M. LaValle, Planning Algorithms, Cambridge.
4. M. Puterman, Markov Decision Processes, Wiley.
5. D. P. Bertsekas, Dynamic Programming and Optimal Control, Athena Scientific.



Prompt Engineering			
Course Code:	CA643	Course Credits:	3
Course Category:	PEC	Course (U / P):	
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Explain the fundamentals of Prompt Engineering and Large Language Models.			
2. Analyse prompt design strategies for diverse AI tasks and applications.			
3. Apply prompt engineering techniques for text, code, image, and multimodal AI systems.			
4. Design intelligent AI applications using Retrieval-Augmented Generation (RAG), function calling, and AI agents.			
5. Evaluate prompt quality considering accuracy, robustness, safety, ethics, and performance.			
COURSE OUTCOMES			
At the end of the course, the students should be able to:			
1. Explain the concepts and architecture of prompt engineering for foundation models.			
2. Analyse prompt optimization techniques for different AI applications.			
3. Apply prompt engineering methodologies for solving real-world problems.			
4. Design AI-powered applications integrating prompts, RAG, APIs, and AI agents.			
5. Evaluate prompt effectiveness using technical, qualitative, and quantitative metrics.			
6. Develop responsible, secure, and scalable prompt-driven AI solutions.			

UNIT I: FOUNDATIONS OF PROMPT ENGINEERING

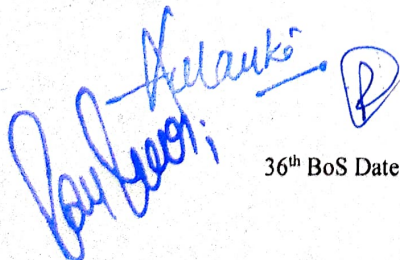
Artificial Intelligence foundations; Generative AI; Foundation Models; Transformer architecture; Tokens and tokenization; Context windows; Embeddings; Prompt lifecycle; Prompt anatomy; Zero-shot learning; One-shot learning; Few-shot learning; Instruction prompting; Role prompting; Chain-of-thought prompting; Self-consistency; Prompt templates; Prompt libraries; Prompt management; Prompt evaluation; AI model capabilities and limitations; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures.

UNIT II: ADVANCED PROMPTING TECHNIQUES

Prompt optimization; Prompt decomposition; Step-by-step reasoning; Tree-of-Thoughts; Graph-of-Thoughts; ReAct prompting; Self-reflection; Prompt chaining; Structured prompting; XML and JSON prompting; Function calling; Tool calling; Context engineering; Long-context prompting; Prompt compression; Prompt versioning; Prompt debugging; Prompt testing; Prompt automation; Comparative evaluation of techniques; Performance measures; Contemporary prompting frameworks; Design trade-offs; Tool-supported exercises.

UNIT III: CONTEXT-AWARE PROMPTING AND RETRIEVAL

Context engineering; Retrieval-Augmented Generation (RAG) fundamentals; Vector representations and semantic search; Document chunking and context selection; Grounding and citation generation; Prompting with retrieved context; Function and tool calling from a prompt-engineering perspective; Context-window management; Prompt-based knowledge retrieval; Evaluation of retrieval-augmented prompts; Practical applications and case studies.



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UNIT IV: MULTIMODAL PROMPT ENGINEERING

Multimodal AI; Vision-language models; Image prompting; Image generation; Image editing prompts; Audio prompting; Video prompting; Code generation; SQL generation; Report generation; Prompt engineering for software development; Prompt engineering for data analytics; Prompt engineering for education; Healthcare AI; Business intelligence; Domain-specific prompt design; Workflow automation; Human-AI collaboration; Comparative evaluation of techniques; Contemporary AI platforms; Industry case studies.

UNIT V: EVALUATION, SAFETY AND ENTERPRISE APPLICATIONS

Prompt evaluation; Benchmarking; Hallucination detection; Robustness testing; Prompt injection attacks; Jailbreak attacks; AI security; Responsible AI; Explainability; Fairness; Bias mitigation; Privacy; Ethical AI; AI governance; Cost optimization; Latency optimization; Model monitoring; PromptOps; Enterprise deployment; AI compliance; Emerging trends in Prompt Engineering; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Contemporary tools and practices.

Text Books / Reference books:

1. Lee Boonstra, Prompt Engineering, O'Reilly Media.
2. Noah Gift, Practical Generative AI with ChatGPT, O'Reilly Media.
3. Chip Huyen, AI Engineering, O'Reilly Media.
4. John Berryman and Albert Ziegler, Prompt Engineering for LLMs, O'Reilly Media.
5. LangChain Documentation (latest edition).
6. OpenAI API Documentation.
7. Anthropic Prompt Engineering Guide.
8. Google Gemini API Documentation.



Evolutionary Computation			
Course Code:	CA645	Course Credits:	3
Course Category:	PEC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. To introduce the main concepts, techniques, and applications in the field of Evolutionary Computation.			
2. To give students practical experience in evolutionary computation techniques and how to implement them using different programming languages.			
3. To develop an understanding of issues in computing using evolutionary algorithms, particularly Genetic Algorithms.			
4. To learn how Evolutionary Computation can be applied to various application areas.			
5. To learn multi-objective optimization.			
COURSE OUTCOMES			
At the end of the course, the students should be able to:			
1. Understand the implementation issues of evolutionary algorithms.			
2. Determine appropriate parameter settings to make different evolutionary algorithms work effectively.			
3. Design new evolutionary operators, representations, and fitness functions for specific practical and scientific applications.			
4. Effectively present an evolutionary computation article to an audience.			
5. Review and critique evolutionary computation articles.			

UNIT I: FUNDAMENTAL CONCEPTS AND DIALECTS

Metaheuristics; Requirements of metaheuristics in optimization problems; Characteristics of problems suitable for application of metaheuristics; Evolutionary Computing metaphor and inspiration from biology; Applications, pros and cons of Evolutionary Computation; Components of Evolutionary Computation; Example applications – Eight Queen Problem, Knapsack Problem; Various dialects of Evolutionary Computation.

UNIT II: GENETIC ALGORITHMS

Introduction; Canonical Genetic Algorithm; Binary, Integer, Real-valued and Permutation representations and variation operators for them; Population models; Parent selection – Fitness proportionate selection, Rank-based selection, Tournament selection; Survivor strategy; Implementation issues; Parameter control and effect on GA dynamics.

UNIT III: THEORETICAL FOUNDATION OF GENETIC ALGORITHMS

Schemas and Hyperplane Sampling; Schemata Theorem; Limitations and Building Block Hypothesis; Two-Armed Bandit Problem; Deceiving a Genetic Algorithm; Minimal Deceptive Problem; Royal Road Functions; SAHC; NAHC; RMHC; Hitchhiking; Exact mathematical models of SGA; Statistical-mechanics approaches.

UNIT IV: PROBLEM SOLVING USING EVOLUTIONARY COMPUTATION

Evolving computer programs; Evolving LISP programs; Evolving Cellular Automata; Data analysis and prediction; Predicting dynamical systems; Predicting protein structure; Evolving Neural Networks – Evolving weights, Architecture (Direct Encoding and Grammatical Encoding), and Learning Rules;

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Baldwin Effect and Evolutionary Reinforced Learning.

UNIT V: PARALLEL IMPLEMENTATION AND OTHER DIALECTS OF EVOLUTIONARY COMPUTATION

Parallel implementation of Genetic Algorithms; Genetic Programming; Evolutionary Programming; Evolutionary Strategies; Learning Classifier Systems; Memetic Algorithms; Introduction to Multi-Objective Genetic Algorithm (MOGA); Introduction to Simulated Annealing; Artificial Bee Colony Algorithms; Ant Colony Optimization Algorithms; Swarm Intelligence; Evolutionary Computation.

Textbooks / Reference books:

1. Melanie Mitchell, An Introduction to Genetic Algorithms, MIT Press.
2. David E. Goldberg, Genetic Algorithms in Search, Optimization and Machine Learning, Pearson Education.
3. Lawrence Davis, Handbook of Genetic Algorithms, Van Nostrand Reinhold.
4. Kalyanmoy Deb, Multi-Objective Optimization Using Evolutionary Algorithms, Wiley.
5. Marco Dorigo and Thomas Stützle, Ant Colony Optimization, MIT Press.

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Generative AI			
Course Code:	CA647	Course Credits:	3
Course Category:	PEC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Explain the principles and terminology of generative models for artificial intelligence			
2. Analyse problems and trade-offs concerning image generation and synthesis.			
3. Apply suitable methods drawn from diffusion models and vision transformers.			
4. Design solutions that integrate multimodal generative vision systems.			
5. Evaluate approaches and responsible practice relating to generative AI applications			
COURSE OUTCOMES			
At the end of the course, the students should be able to:			
1. Explain the fundamental concepts and terminology of generative models for artificial intelligence			
2. Analyse problems using the methods covered under image generation and synthesis.			
3. Apply appropriate techniques from diffusion models and vision transformers to realistic vision applications.			
4. Design a solution that integrates concepts from multimodal generative vision systems.			
5. Evaluate alternative generative AI approaches using technical, performance and reliability criteria.			
6. Develop and communicate a responsible, reproducible solution drawing on generative AI			

UNIT I: FOUNDATIONS OF GENERATIVE AI

Generative AI fundamentals; Probabilistic generative models; Latent representations; Image distributions; GANs; VAEs; Diffusion models; Transformer-based vision models; Datasets; Image generation pipeline; Evaluation metrics; Applications; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations; Simulation-based exercises.

UNIT II: IMAGE GENERATION AND SYNTHESIS

Image synthesis; GAN architectures; Conditional GANs; StyleGAN; Image-to-image translation; Image super-resolution; Image inpainting; Image colorization; Style transfer; Controllable image editing; Quality assessment; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations; Simulation-based exercises; Hardware deployment issues.

UNIT III: DIFFUSION MODELS AND TRANSFORMERS

Diffusion probabilistic models; Denoising diffusion; Latent diffusion; Stable Diffusion; Text-to-image generation; Prompt conditioning; Vision Transformers; Attention mechanisms; Controllable generation; Model fine-tuning; Performance optimization; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations; Simulation-based exercises; Reproducibility practices.

UNIT IV: MULTIMODAL GENERATIVE VISION

Vision-language models; CLIP; Image captioning; Visual question answering; Text-guided image editing; Image retrieval; Multimodal embeddings; Cross-modal learning; Synthetic data generation; Evaluation and benchmarking; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations; Simulation-based exercises.

UNIT V: APPLICATIONS AND RESPONSIBLE GENERATIVE AI

Medical image generation; Autonomous driving; Industrial inspection; Robotics; Digital content creation; Synthetic datasets; Deployment considerations; Model compression; Explainability; Fairness; Bias; Copyright and intellectual property; Privacy; Responsible AI; Emerging trends; Worked examples and case analysis; Comparative evaluation of techniques; Implementation considerations; Performance and quality measures; Common limitations and failure scenarios; Contemporary tools and practices; Safety and robustness considerations; Ethical and societal implications.

Textbooks / Reference books:

1. David Foster, Generative Deep Learning, 2nd ed., O'Reilly Media.
2. Ian Goodfellow, Yoshua Bengio and Aaron Courville, Deep Learning, MIT Press.
3. Richard Szeliski, Computer Vision: Algorithms and Applications, 2nd ed., Springer.
4. Rafael C. Gonzalez and Richard E. Woods, Digital Image Processing, 4th ed., Pearson.
5. Ashish Vaswani et al., Attention Is All You Need, NeurIPS Proceedings.
6. C. Blum and D. Merkle, Swarm Intelligence: Introduction and Applications, Springer.



Predictive Analytics			
Course Code:	CA649	Course Credits:	3
Course Category:	PEC	Course (U / P):	P
Course Year (U / P):	2P	Course Semester (U / P):	3P
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Understand the concepts and techniques of predictive analytics.			
2. Apply statistical and machine learning methods for predictive modeling.			
3. Analyse structured and unstructured data to extract predictive insights.			
4. Develop predictive models for real-world decision-making.			
5. Evaluate predictive models using appropriate performance metrics and ethical considerations.			
COURSE OUTCOMES			
At the end of the course, the students should be able to:			
1. Explain the principles and applications of predictive analytics.			
2. Apply statistical and machine learning techniques for prediction.			
3. Analyse datasets to identify patterns and predictive insights.			
4. Develop predictive models for real-world applications.			
5. Evaluate predictive models using appropriate statistical and machine learning performance metrics.			
6. Assess predictive models in deployment with respect to reliability, data drift, fairness, privacy, security, and responsible AI considerations.			

UNIT I: INTRODUCTION TO PREDICTIVE ANALYTICS

Introduction to Predictive Analytics; Predictive, Descriptive and Prescriptive Analytics; Data Analytics Life Cycle; Data Collection and Integration; Data Preprocessing; Data Cleaning; Feature Engineering; Exploratory Data Analysis (EDA); Data Visualization; Statistical Foundations; Probability Distributions; Sampling Techniques; Business Applications of Predictive Analytics.

UNIT II: STATISTICAL AND MACHINE LEARNING TECHNIQUES

Regression Analysis; Linear and Logistic Regression; Decision Trees; Random Forest; Naïve Bayes; K-Nearest Neighbours (KNN); Support Vector Machines (SVM); Ensemble Learning; Model Selection; Cross Validation; Bias-Variance Tradeoff; Hyperparameter Tuning; Feature Selection.

UNIT III: TIME SERIES AND FORECASTING

Introduction to Time Series Data; Trend and Seasonality; Time Series Decomposition; Moving Average Models; Exponential Smoothing; ARIMA Models; Forecast Accuracy Measures; Demand Forecasting; Financial Forecasting; Sales Prediction; Weather Prediction; Predictive Maintenance; Anomaly Detection.

UNIT IV: DEEP LEARNING AND ADVANCED PREDICTIVE ANALYTICS

Neural Networks; Deep Learning Fundamentals; Artificial Neural Networks (ANN); Recurrent Neural Networks (RNN); Long Short-Term Memory (LSTM); Predictive Analytics using Deep Learning; Natural Language Processing for Prediction; Sentiment Analysis; Recommendation Systems; Predictive Analytics using Big Data; AutoML; Explainable AI (XAI).

UNIT V: DEPLOYMENT, ETHICS AND INDUSTRY APPLICATIONS

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Model Evaluation Metrics; Accuracy, Precision, Recall, F1-Score, ROC-AUC; Model Deployment; MLOps Fundamentals; Model Monitoring; Data Drift and Concept Drift; Responsible AI; Fairness and Bias in Predictive Models; Privacy and Security; Cloud-Based Predictive Analytics; Industry Case Studies in Healthcare, Finance, Retail, Manufacturing & Smart Cities; Future Trends.

Textbooks / Reference books:

1. Eric Siegel, Predictive Analytics: The Power to Predict Who Will Click, Buy, Lie, or Die, Wiley.
2. Dean Abbott, Applied Predictive Analytics, Wiley.
3. Trevor Hastie, Robert Tibshirani and Jerome Friedman, The Elements of Statistical Learning, Springer.
4. Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani, An Introduction to Statistical Learning, Springer.
5. Jiawei Han, Micheline Kamber and Jian Pei, Data Mining: Concepts and Techniques, Morgan Kaufmann.
6. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow, O'Reilly Media.
7. Ian Goodfellow, Yoshua Bengio and Aaron Courville, Deep Learning, MIT Press.

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H. Lakshmi, Jayaraman P



SEMESTER IV

CA692: Dissertation Part-II

Adarsh Varma 